TRENESIS

# NUTRITION BASICS

Evidence-based fundamentals for building real, lasting results.

NO FLUFF · NO PSEUDOSCIENCE · JUST RESULTS

BEFORE YOU START

# Disclaimer & How to Use This Guide

This guide is for educational purposes only and is based on general nutrition principles. It is not intended as medical advice or a substitute for professional healthcare guidance. If you have specific health conditions or dietary needs, consult a qualified healthcare professional.

The information presented is based on current evidence-based nutrition science. Individual results vary based on consistency, training, sleep, and genetics.

## How to Use This Guide

- Read Chapters 1–2 first. Understanding calories and protein unlocks everything else.
- Use the macro targets in Chapter 3 as a starting point – adjust based on results.
- Supplements (Chapter 4) are optional extras, not foundations.
- Combine with the Trenesis Strength Training Guide for best results.

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CHAPTER 01

# Calories & Energy Balance

The one concept that governs body composition.

## What Is a Calorie?

A calorie is simply a unit of energy. Your body requires energy to breathe, move, think, digest food, and repair tissue. Everything you eat provides some amount of energy; everything you do burns some amount of energy. The relationship between these two numbers determines whether your body weight goes up, down, or stays the same.

## Energy Balance

Energy balance is the cornerstone of nutrition science. It is not the whole story – food quality, hormones, and micronutrients all matter – but it is the dominant factor for body composition:

| STATE | CALORIES vs. TDEE | RESULT |
|---|---|---|
| **Deficit** | Less than TDEE | Weight Loss |
| **Maintenance** | Equal to TDEE | Weight Stable |
| **Surplus** | More than TDEE | Weight Gain |

## Finding Your TDEE

TDEE stands for Total Daily Energy Expenditure – the total calories your body burns in a day, including exercise. A rough starting point:

| ACTIVITY LEVEL | DAILY CALORIES (estimate) |
|---|---|
| **Sedentary (desk job, little exercise)** | Bodyweight (kg) × 28–31 |
| **Moderately active (3–4 sessions/week)** | Bodyweight (kg) × 32–35 |
| **Very active (5+ sessions/week)** | Bodyweight (kg) × 36–40 |

You can also use a TDEE calculator if you don't want to do the math.

These are estimates only. It can be helpful to track your calorie intake and body weight for a couple of weeks so you can see how your body actually responds and fine-tune things from there. The scale is your feedback mechanism.

> **KEY INSIGHT**
>
> A deficit or surplus of roughly 500 kcal/day produces approximately 0.5 kg of weight change per week. Start conservatively – aggressive cuts sacrifice muscle; aggressive surpluses add unnecessary fat.

## Do You Need to Bulk?

You can build muscle without a large calorie surplus. In fact, eating far above maintenance is generally not recommended, as it doesn't significantly accelerate muscle growth and mostly leads to increased fat gain.

Beginners–and people returning to training after a break–can often gain muscle at maintenance calories, and in some cases even with a slight deficit.

For most trainees, a small surplus of around 200–300 kcal per day is typically sufficient to support muscle growth while keeping fat gain minimal.

> **THE TAKEAWAY**
>
> Eat slightly above maintenance (+200–300 kcal) when your primary goal is muscle gain. Eat slightly below maintenance (−300–500 kcal) when your primary goal is fat loss. Avoid extremes in either direction.

CHAPTER 02

# Protein – The Priority Macro

Get this right and most other things fall into place.

## Why Protein Is Different

Protein is the only macronutrient that directly provides the building blocks for muscle tissue. Amino acids – the components of dietary protein – are used to repair muscle fibres damaged during training, build new muscle, support immune function, produce enzymes and hormones, and maintain connective tissue.

Unlike carbohydrates and fat, the body has no dedicated protein storage. Any excess is oxidised for energy or converted to fat. This means consistent daily protein intake matters far more than sporadic high-protein days.

## How Much Protein Do You Need?

The commonly cited 0.8 g/kg bodyweight is more of a minimum to prevent deficiency than an optimal target for muscle growth. For active individuals, it's often too low.

A more effective range is about 1.5–2.2 g/kg or even slightly above.

> **MYTH**
>
> A common myth is that high protein intake harms kidneys. In people with normal kidney function, higher protein does not cause damage. The main downside of very high protein is that it can crowd out carbs and fats in the diet, which are also important for energy and overall health.

## Protein Timing

The "anabolic window," where you supposedly need to eat a lot of protein immediately after training, isn't strongly supported by evidence. Muscle growth depends more on your total daily protein intake than on precise timing. Spreading protein evenly across the day could still help support muscle protein synthesis and recovery.

> **KEY INSIGHT**
>
> Protein has the highest thermic effect of any macronutrient – your body burns roughly 20–30% of its calories just through digestion. High-protein diets also tend to improve satiety, making it easier to sustain a calorie deficit.

CHAPTER 03

# The Other Macronutrients

Carbohydrates and fat both have a role – neither is the enemy.

Once you hit your calorie and protein targets, the split between carbohydrates and fat is largely a matter of personal preference, food tolerance, and lifestyle. Both macros serve important physiological functions.

## Carbohydrates

Carbohydrates are the body's preferred fuel source during high-intensity exercise. Muscle glycogen – the stored form of carbohydrate – directly powers your training sets. Low glycogen impairs performance, reduces training volume, and can increase muscle breakdown. Consuming carbohydrates a few hours before training can help improve performance.

## Fat

Dietary fat is essential for hormone production (including testosterone), fat-soluble vitamin absorption (A, D, E, K), cell membrane integrity, and joint health. It is calorie-dense (9 kcal/g vs 4 kcal/g for protein and carbs), so it needs to be accounted for carefully in a deficit. Prioritise unsaturated fats(olive oil, avocado, nuts, seeds, and fatty fish) while keeping saturated fat (meat, dairy) in moderation.

## Putting It Together

A simple hierarchy for setting up your nutrition:

| STEP | ACTION |
|---|---|
| **1 – Calories** | Set based on your goal (deficit / maintenance / surplus) |
| **2 – Protein** | Hit 1.5–2.2 g/kg bodyweight every day |
| **3 – Carbs** | Try to hit at least 3–5 g/kg bodyweight every day |
| **4 – Fat** | Fill remaining calories with fat (≈ 0.6–1 g/kg bodyweight every day) |

CHAPTER 04

# Supplements

Can support gym progress.

The supplement industry is a multi-billion-euro business built largely on marketing. The vast majority of products on the shelf are not necessary for most people.

It's important to understand that supplements are only a small, optional addition–they won't meaningfully improve your results if your training, nutrition, sleep, and overall consistency aren't already in place.

For people who strength train, there are a few supplements with strong, consistent evidence behind them, with protein powder and creatine among the most well-established and widely supported.

## Protein Powder

Protein powder is not a special anabolic substance – it's simply a convenient way to get protein in your diet. Whey protein (from milk) is the most common but plant-based protein powders (like pea and rice blends) can be just as effective.

- Mix with water or milk to make a shake, or add to foods like baked goods.
- Prioritize a balanced diet: protein powder cannot replace the nutritional value of whole foods.
- A typical serving provides about 20–30 g of protein for roughly 100–130 kcal.

## Creatine Monohydrate

Creatine is the single most studied supplement in sports science and one of the very few with unambiguous performance benefits. It works by increasing phosphocreatine stores in muscle, which helps regenerate ATP during short, high-intensity efforts – exactly what strength training demands.

- The standard dose is 3–5 g per day (or 1g/10kg bodyweight). No loading phase is necessary.
- Take it at any time – timing does not affect results.
- A 1–2 kg increase in body weight in the first 1–2 weeks (water retention in muscle) is possible. This is normal.
- Benefits include: slightly more reps at a given weight, improved recovery between sets, and.
- Creatine monohydrate is the cheapest and most effective form. Ignore expensive variants.
- Some research even suggests small improvements in certain cognitive tasks.

### KEY INSIGHT

Supplements sit at the bottom of the hierarchy. No supplement compensates for a poor diet. Fix nutrition first.

### BUY SUPPLEMENTS

If you're looking to buy protein powder or creatine monohydrate, check out our favorite **Supplements** available on our website.

CHAPTER 05

# Frequently Asked Questions

### Do I need to count calories?

Not necessarily – but awareness matters. If you are not getting results, tracking calories for even 2–4 weeks creates invaluable baseline knowledge. Many people significantly underestimate their intake until they measure it.

### How important is meal timing?

Less important than total daily intake. Eat enough protein, spread across 3–5 meals, and hit your calorie target. Whether you eat breakfast or fast until noon matters far less than what you eat overall.

### Can I build muscle as a vegetarian or vegan?

Yes – absolutely.

You can build muscle on a vegetarian or vegan diet as long as your nutrition is well planned. The main thing is meeting your protein needs, since plant-based sources are sometimes less protein-dense.

With a good mix of plant proteins (and protein powder if needed), it works just as well for muscle growth as any other diet.

### Does soy lower testosterone?

No. Soy contains phytoestrogens, which are structurally similar to human estrogen but don't act like it in the body. By the way: the estrogen naturally present in cow's milk is more similar to human estrogen, but neither soy nor milk has been shown to meaningfully affect hormone levels in typical dietary amounts.

### What about rest days – should I eat less?

Slightly less is fine, but drastic cuts on rest days are unnecessary. Muscle repair and recovery happen primarily on rest days, so your body still needs adequate protein and calories.

### Are carbs bad for you?

No. Carbohydrate quality matters – vegetables, whole grains, and fruit are preferable to ultra-processed options – but carbohydrates are not inherently fattening. Excess calories cause fat gain, regardless of macronutrient source.

## Ready to put it into practice?

More free guides, training content, and resources at:

**trenesisgym.github.io/trenesis/**

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